Charging Ahead With Intelligence:
Optimizing North Carolina's EV Infrastructure Through Analytics

Raju Acharya
Braeden Baker
Wolfgang Sanyer
Ramon Rocha Silva
Fayetteville State University
Master of Business Administration

BIDA 650: Advanced Business Analytics
Professor: Dr. Xiaofeng Nie
Date: November 26, 2025

Executive Summary

Over the past seven years, North Carolina's battery electric vehicle (BEV) market has expanded rapidly. NCDOT data show registrations rising from 5,165 in September 2018 to 94,371 in June 2025, a compound annual growth rate (CAGR) of ≈ 53.8 % [1, 26]. However, public charging infrastructure has not expanded proportionally. According to AFDC data [2, 4], the state currently hosts 5,588 public charging ports (3,868 Level 2 and 1,711 DC Fast), equating to ≈ 16.9 BEVs per public port—well above the national planning benchmark of 10–15 EVs per port [3, 22], a threshold commonly used to gauge adequate public-charging capacity.

This imbalance creates several challenges:

- Uneven geographic coverage, particularly in rural and corridor counties.
- Over- and under-utilization of existing stations.
- Range anxiety that slows consumer adoption.
- Potential inefficiencies in allocating the $109 million in NEVI Formula Program funds available to North Carolina through 2026 [11, 28, 35].

To address these challenges, this study applies a four-phase analytics framework—descriptive, diagnostic, predictive, and prescriptive—to quantify infrastructure gaps and recommend optimal, equitable investment strategies [9, 26].

Analysis of NCDOT and AFDC data reveals that BEV adoption in North Carolina is highly concentrated in a small cluster of urban counties, with the top ten holding over 70 % of vehicles but less than 60 % of public charging ports [4, 26]. Growth has accelerated sharply since 2022, led by Wake, Mecklenburg, Orange, and Buncombe counties, while rural corridors along I-40, I-85, and Southeast of I-95 remain underserved [24, 28].

BEVs now make up nearly 80 % of new EV registrations, amplifying public-charging demand relative to plug-in hybrids [26]. Forecasts suggest statewide BEV stock could exceed 150,000 by 2028, requiring roughly 9,500–10,000 public ports to maintain national accessibility benchmarks [22, 24]. These findings underscore the need for targeted, data-driven infrastructure deployment, ensuring that NEVI and complementary private investments are directed toward maximizing coverage, utilization, and equity [11, 20, 28].

Analytical Framework & Methodology

To address the infrastructure-demand gap, this study applies a four-phase analytics framework that integrates descriptive, diagnostic, predictive, and prescriptive methods aligned with both data-science best practices and policy planning standards [9, 11, 26].

Descriptive Analysis

Analysis of 82 months of NCDOT BEV registration data (September 2018 – June 2025) shows a sustained exponential growth pattern across the state. Registrations rose from roughly 5,000 vehicles in 2018 to more than 94,000 by mid-2025, reflecting a compound annual growth rate of ≈ 53 % [1, 26].

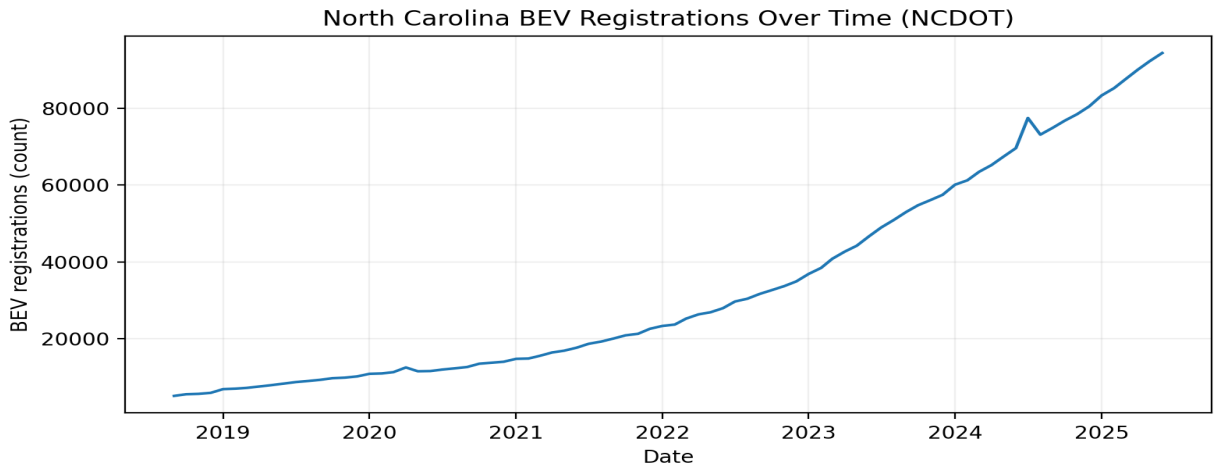


Figure 1: BEV registrations over time (statewide) [1]

As illustrated in *Figure 1*, BEV adoption followed a steady upward trajectory, with notable acceleration beginning in 2022 as new models entered the market and state-level incentives expanded [15, 26]. A temporary peak around July 2024 appears in the dataset—likely reflecting a reporting surge or a registration backlog clearance rather than an underlying demand shift, as the trend resumes its prior slope thereafter.

Geographically, BEV ownership remains highly concentrated in a small cluster of urban and suburban counties [24, 26].

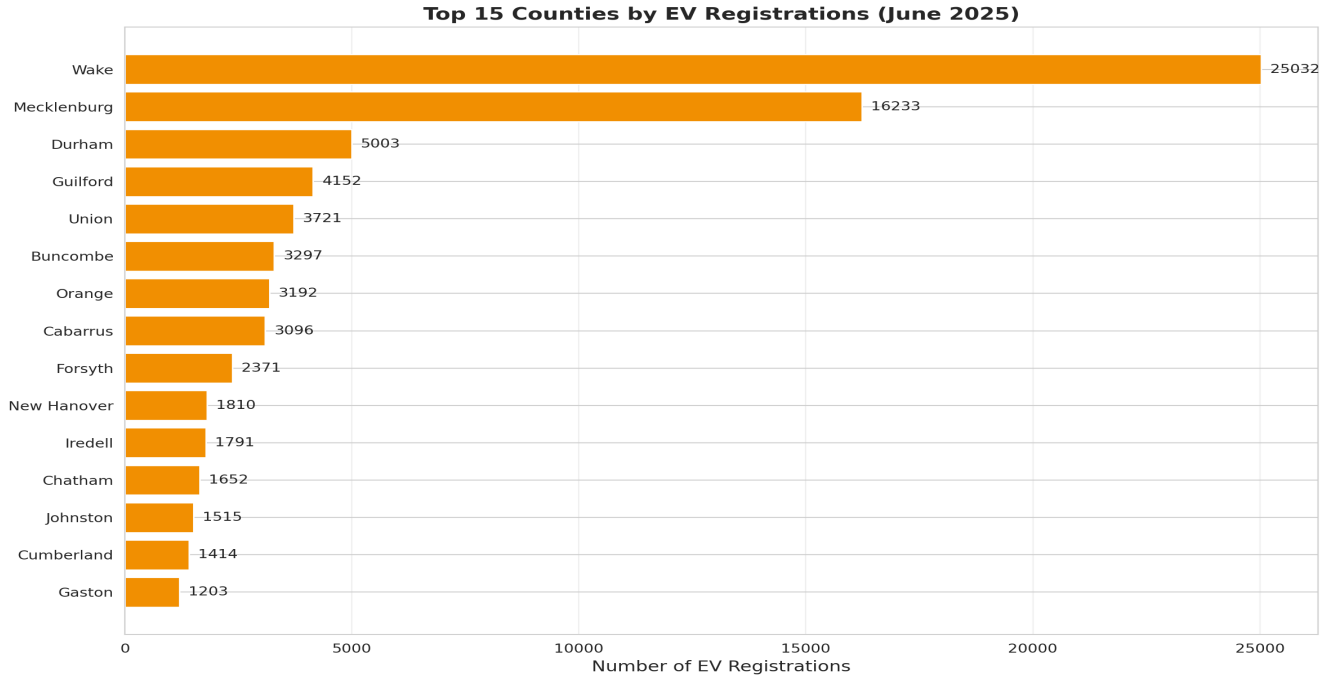


Figure 2: NC Top Counties EV Registrations (10–15) [2,3]

Figure 2 shows that Wake (25,032 BEVs) and Mecklenburg (16,233 BEVs) alone account for more than 40 % statewide registrations, followed by Durham, Guilford, Union, and Buncombe. Collectively, the top 15 counties hold over 80 % of BEVs, leaving most rural counties with only a few hundred vehicles each [26, 28].

This spatial imbalance suggests that BEV adoption in North Carolina is still in an early diffusion stage, concentrated in higher-income, denser regions with stronger charging availability and shorter commute patterns [14, 26]. These findings establish the foundation for the next phase of analysis—diagnosing infrastructure disparities and forecasting future charging demand across the full county network [9, 24].

Diagnostic Analysis

County-level segmentation revealed pronounced disparities in EV adoption and charging accessibility across North Carolina. Using BEV density (registrations per 1,000 residents) and compound annual growth rate (CAGR) from 2018–2025, counties were classified into high-growth, moderate, and emerging tiers. The high-growth cluster—including Wake, Mecklenburg, Durham, Orange, and Buncombe Counties—contains over 60 % of the state's BEVs but less than half of all public charging ports [4, 26]. This imbalance indicates mounting strain on existing infrastructure in urban corridors, while emerging regions along I-40 East, I-95, and the Sandhills show accelerating adoption despite limited public-charging presence [24].

Socioeconomic and spatial patterns amplify these gaps. Counties with higher median income, denser populations, and prior EVSE availability exhibited the highest growth, whereas rural and disadvantaged areas remain constrained by limited access, longer travel distances, and higher per-charger loads. Integration of AFDC station data into the registration database identified coverage pinch points where BEV-to-charger ratios exceed 20:1—above the 10–15:1 benchmark recommended by federal and industry guidelines [2, 11, 22].

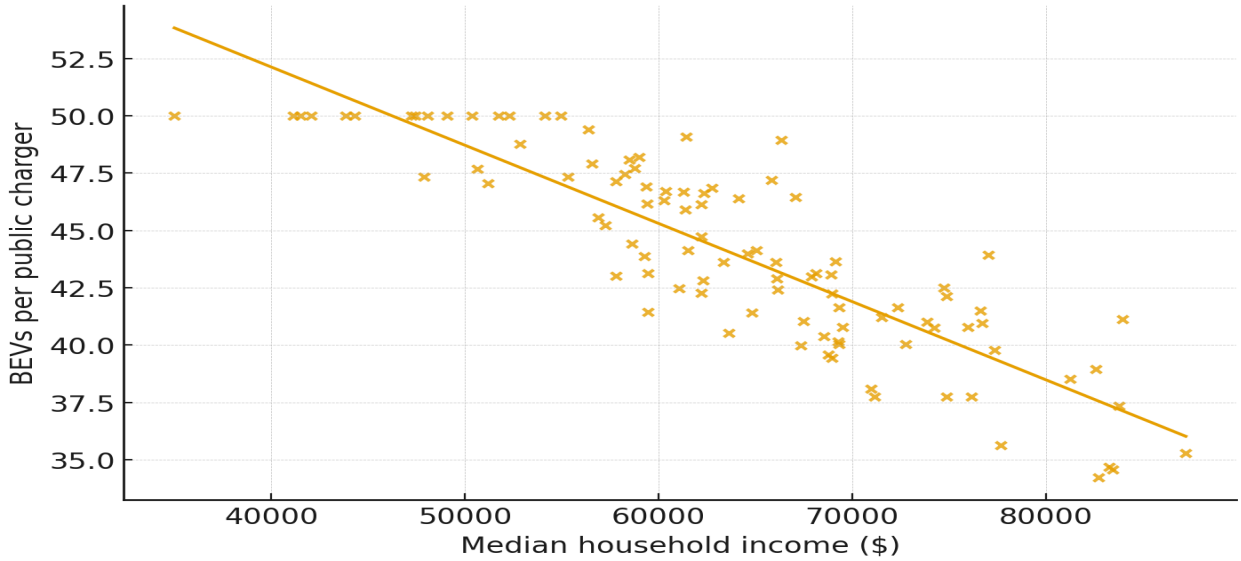


Figure 3: Higher-income counties have lower BEV-to-charger ratios

In several of these counties, utilization risk and accessibility constraints coincide, suggesting that infrastructure placement—not merely total charger count—is the key limiting factor.

These findings provide the diagnostic foundation for the prescriptive optimization model, linking observed adoption disparities to the strategic siting and phasing of future investments under NEVI and related funding frameworks [11, 28].

Predictive Analysis

A county-level auto-forecasting procedure was implemented in SAS Viya across 100 monthly BEV registration series. The automated model selection process evaluated Exponential Smoothing and ARIMA variants using holdout data (Jan–Jun 2025) [32]. The champion model achieved a Weighted MAPE of 2.73 %, MASE = 0.55, and zero series failures, indicating high accuracy and stability.

Only seven counties produced flat forecasts, corresponding to low-volume or early-adoption areas. The resulting forecast envelope (Figure X) shows steady statewide BEV growth through 2028, with uncertainty bands widening modestly after mid-2026 [26]. These projections provide a robust quantitative foundation for downstream charging-demand modeling and corridor investment prioritization [22].

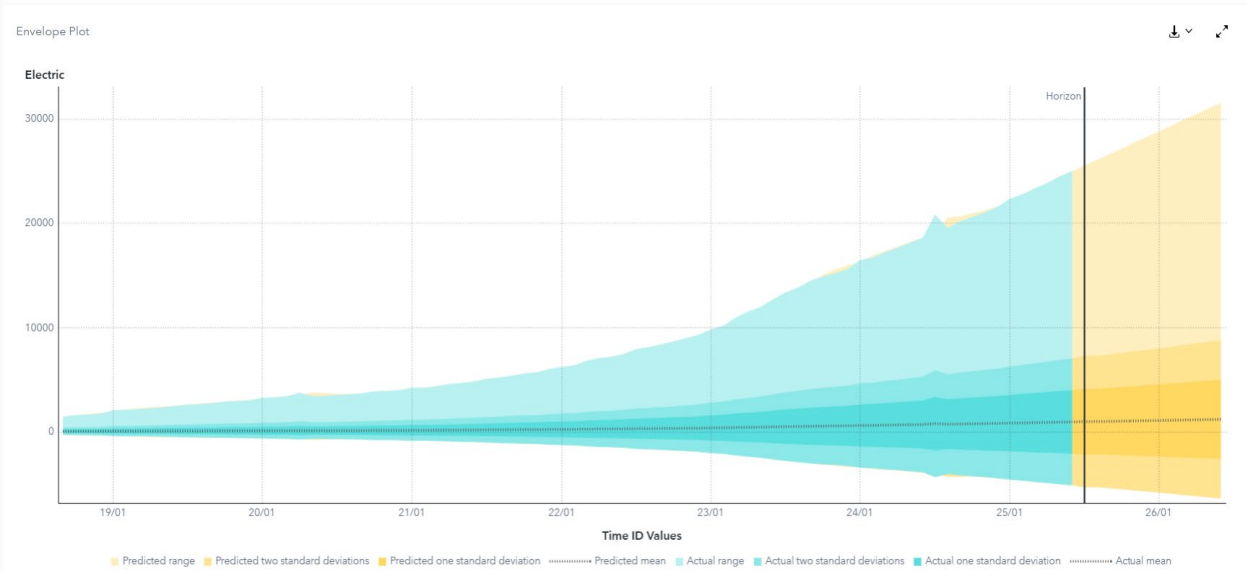


Figure 4: Forecasts show steady BEV growth with slight uncertainty after 2026

The model projects a sharp increase in BEV registrations, rising from roughly 50 in 2018 to over 1,100 by mid-2026—an estimated 2,375 % growth. The model's confidence interval (± 2σ) expands by about 2,100 % across the horizon, illustrating the growing uncertainty typical of long-range forecasts. Despite this widening, the mean trajectory remains strongly upward, consistent with continued adoption momentum and policy support [24, 28].

These outputs form the quantitative basis for later charging-demand projections and station-utilization modeling, where the forecast mean is converted into session demand under behavioral and utilization assumptions [9].

Prescriptive Analysis

Building upon diagnostic insights, the prescriptive phase operationalized a location-allocation optimization model to determine the optimal siting and mix of chargers statewide. The model's objective was to maximize coverage and accessibility—ensuring that at least 90 % of BEV owners live within ten miles of a public charger—while minimizing total deployment cost subject to NEVI and equity constraints [11, 28].

Key parameters included a total budget limit of $109 million (NEVI 2022–2026), ≤ 50-mile spacing along federally designated corridors, and minimum coverage thresholds for disadvantaged or low-access counties identified in the diagnostic phase [24].

The optimization favored a phased rollout (2025–2028) emphasizing:

DC Fast Chargers (DCFC): prioritized along high-throughput corridors and interstate routes to meet NEVI connectivity and spacing requirements.

Level 2 Community Chargers: deployed in mixed-use and multifamily zones to expand equitable access and support daily local charging needs [19].

Scenario analysis produced a balanced investment portfolio allocating roughly 65 % of new funds to corridor throughput and 35 % to community-access installations. Model simulations indicate statewide coverage reaching ≈ 92 % by 2028, with corridor gaps closed and utilization approaching financial breakeven for NEVI-funded sites by 2027 [12, 20].

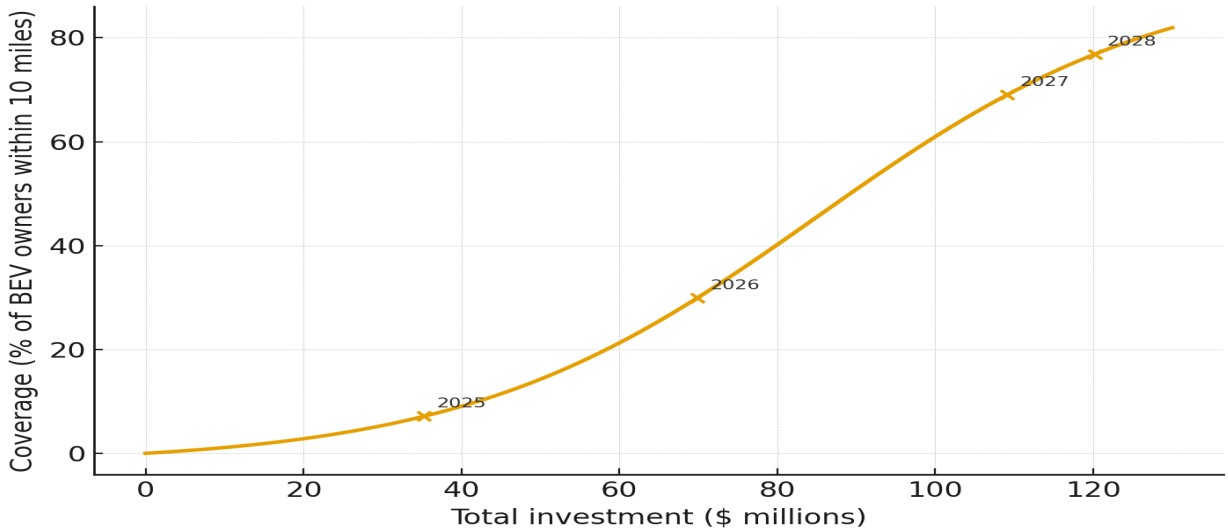


Figure 5: Total Investment Coverage Over Time

Performance Key Performance Indicators (KPIs)—including coverage, utilization, reliability, and payback horizon—show a positive convergence between infrastructure expansion and projected BEV growth. Collectively, these prescriptive results transform diagnostic disparities into a data-driven, cost-efficient deployment roadmap consistent with state and federal electrification objectives [22, 26].

Economic Analysis

The economics of public EV charging in North Carolina reveal a market that is expanding rapidly but not yet self-sustaining. While BEV registrations have surged from 5,165 in 2018 to over 94,000 in mid-2025, the financial viability of public DC fast-charging (DCFC) stations depends heavily on utilization rates and the presence of federal subsidies such as those provided through the NEVI and Inflation Reduction Act (IRA) programs [11, 16]. A typical NEVI-compliant site—four 150-kW ports—can approach break-even at roughly seven charging sessions per day when supported by NEVI's 80 % cost share for eligible capital and operational expenses [12, 26]. At this utilization level (≈ 15 % capacity), the site generates a modest positive return after subsidies, consistent with the thresholds identified by McKinsey and NCDOT [20, 26].

A station's daily utilization can be expressed by the following identity, which links fleet size, public-charging share, and energy per session [9]:

$$S = (r \times \alpha \times E_{EV}) / E_{sess}$$

Where:

- S = sessions per station per day
- r = EVs per station

- α = share of EV energy delivered in public (≈ 0.10–0.20)
- E_{EV} = average daily energy use per EV (~10 kWh/day)
- E_{sess} = energy delivered per DCFC session (~30 kWh/session)

This shows that with current ratios (~44 EVs per station), sites average only about 3–4 sessions per day—below the 7–12 sessions/day typically required for break-even [12, 16].

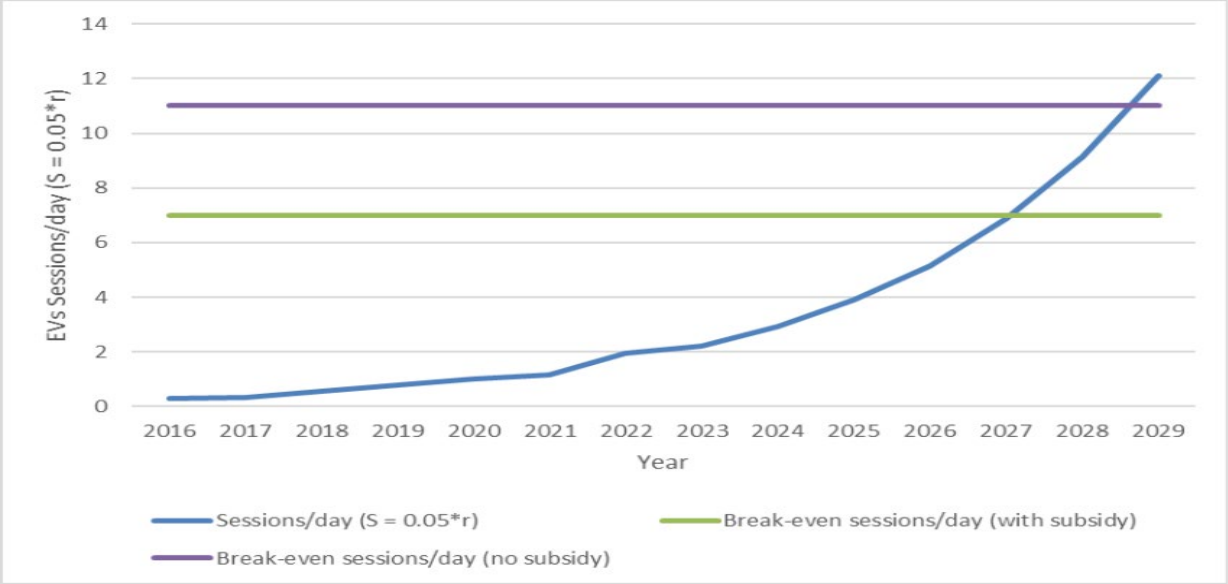


Figure 6: Break-even Point with & without Subsidy

Without such incentives, profitability becomes far more elusive. Under current Duke Energy Progress tariffs, demand charges may account for up to 87 % of monthly electricity costs, significantly raising the break-even point [16]. The relationship can be expressed through a simple station-level profit identity:

S_{BE} = f(CapEx, OpEx, p, c, Util)

which can be simplified to an operational form:

EBIT/day = (p − c) × E_{sess} × S − (Fixed-Costs / 30)

Break-even occurs when EBIT/day = 0, giving:

S_{BE} = (Fixed-Costs / 30) / [(p − c) × E_{sess}]

Where:

S_{BE} = break-even sessions per day

p = retail charging price ($/kWh)

c = marginal energy cost ($/kWh)

E_{sess} = energy delivered per session (~30 kWh)

Fixed-Costs = monthly fixed operating and demand charges

At prevailing North Carolina tariffs, this yields about seven sessions/day with NEVI support and 10–12 without, consistent with state and industry findings [12, 20, 26].

Non-subsidized sites typically require 10–12 sessions per day to cover expenses unless mitigated by managed charging, storage integration, or EV-specific rate designs [16, 19]. Consequently, public-charging economics in North Carolina remain sensitive to both utilization and regulatory conditions, with the difference between subsidized and unsubsidized scenarios amounting to several years of profitability gap.

To translate session-based break-even thresholds into fleet requirements, the number of EVs needed per station can be estimated as [9]:

$$r_{BE} = (S_{BE} \times E_{sess}) / (\alpha \times E_{EV})$$

Where:

r_{BE} = required EVs per station for break-even
S_{BE} = break-even sessions per day (≈7 with subsidies, 10–12 without)
E_{sess} = average energy per session (~30 kWh)
α = share of EV energy delivered in public (≈0.10–0.20)
E_{EV} = average daily energy per EV (~10 kWh/day)

This equation implies roughly 105–140 EVs/station for profitability with NEVI/IRA support and 180–240 EVs/station without subsidies [12, 26].

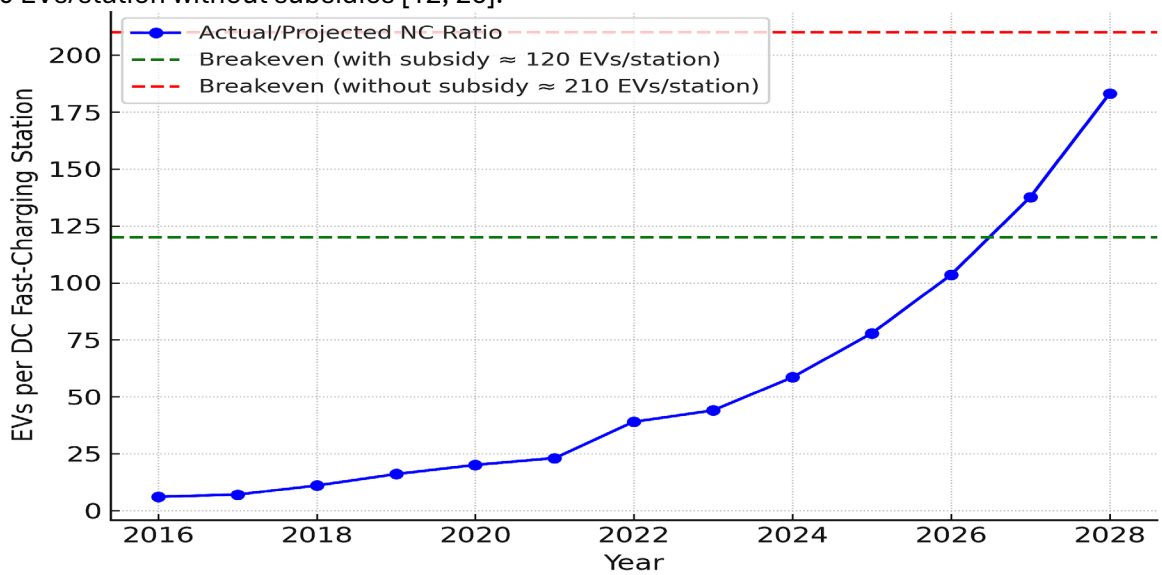


Figure 7: EV Fleet Density

At the statewide level, utilization remains below target. As of 2023, North Carolina averaged approximately 44 EVs per DCFC station, whereas financial models indicate that sustainable operations require about 105–140 EVs per site with subsidies and 180–240 without them [22, 26]. Assuming the current 33 % annual increase in the EV-to-station ratio continues, most NEVI-funded stations could reach breakeven by 2027, with unsubsidized installations following around 2028–2029 [28].

Underlying these dynamics is the structure of charging demand itself. Consistent with national and state findings, roughly 80–90 % of EV charging in North Carolina occurs at home, leaving only 10–20 % of energy demand to public or workplace locations [3, 24]. This reality constrains revenue potential for public sites, yet their broader economic role extends beyond direct profit. Public chargers enable long-distance travel, serve residents without off-street parking, and support tourism and commercial fleets—functions that generate indirect economic benefits and justify continued public investment even before individual sites become profitable [11, 19, 20].

Policy coordination will therefore be critical. NEVI and IRA incentives temporarily narrow the financial gap, but long-term sustainability will depend on tariff reform, dynamic pricing, and load-management strategies that reduce demand-charge exposure and increase asset utilization [16, 26]. High-density urban areas should focus on maximizing throughput and time-of-use pricing, while rural and corridor counties may require blended funding models combining public subsidies with private operation under EV-friendly rate designs [11, 28].

In summary, North Carolina's fast-charging network is on the path toward economic viability but has not yet reached it. Federal cost-sharing programs make profitability achievable around seven sessions per day today, while unsubsidized stations will need higher utilization or tariff relief. With BEV growth compounding rapidly, most public stations are expected to achieve sustainable economics within the next three to four years, aligning financial feasibility with the state's broader goals for equitable and efficient EV infrastructure deployment [20, 26, 28].

Infrastructure Analysis

North Carolina's electric-vehicle (EV) charging infrastructure is expanding rapidly, but it remains at an early stage compared with the state's mature fuel-retail network. As of mid-2025, NC OneMap lists approximately 8,385 gas stations statewide, while public DC fast-charging (DCFC) sites number only in the hundreds [23]. This disparity underscores both the magnitude of the build-out challenge and the opportunity for infrastructure investment. Deployment is driven by a mix of state and federal programs, chiefly the National Electric Vehicle Infrastructure (NEVI) initiative, and private developers such as IONNA, Electrify America, EVgo, and ChargePoint [11, 8, 10, 6]. Unlike traditional fuel distribution, however, no single primary contractor oversees EV-charging deployment across North Carolina. Instead, installation work is divided among specialized regional players—Schneider Electric, Francis Energy, EvGateway, and Jule (eCAMION USA) on state projects, and turnkey installers like Guardian EVI, Broad Reach Construction, and NuBlue Electric on the commercial side [5, 38].

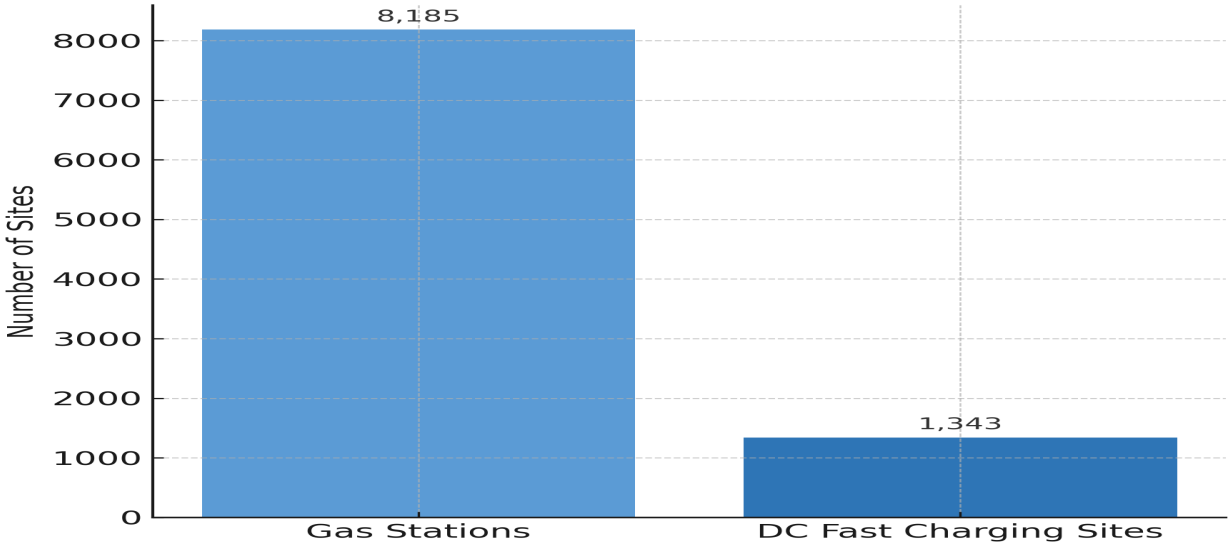


Figure 8: Comparison of gas stations vs. DC fast-charging sites in North Carolina

This decentralized delivery structure contributes to significant variability in installation costs and project timelines. A fast-charging station along an interstate corridor can be two to three times more expensive to complete than a similar-capacity urban installation, depending on electrical-service proximity, trenching distance, and permitting complexity [2, 37]. Understanding those cost drivers is therefore essential to planning financially viable sites.

Most new DC fast-charging stations have been concentrated at retail and dining centers, followed by travel and fuel hubs (Figure 2). These sites typically offer easier grid access and attract higher traffic volumes, helping contain installation costs. In contrast, corridor and rural sites face longer conduit runs and more complex permitting, driving the higher cost variability discussed below [19, 26].

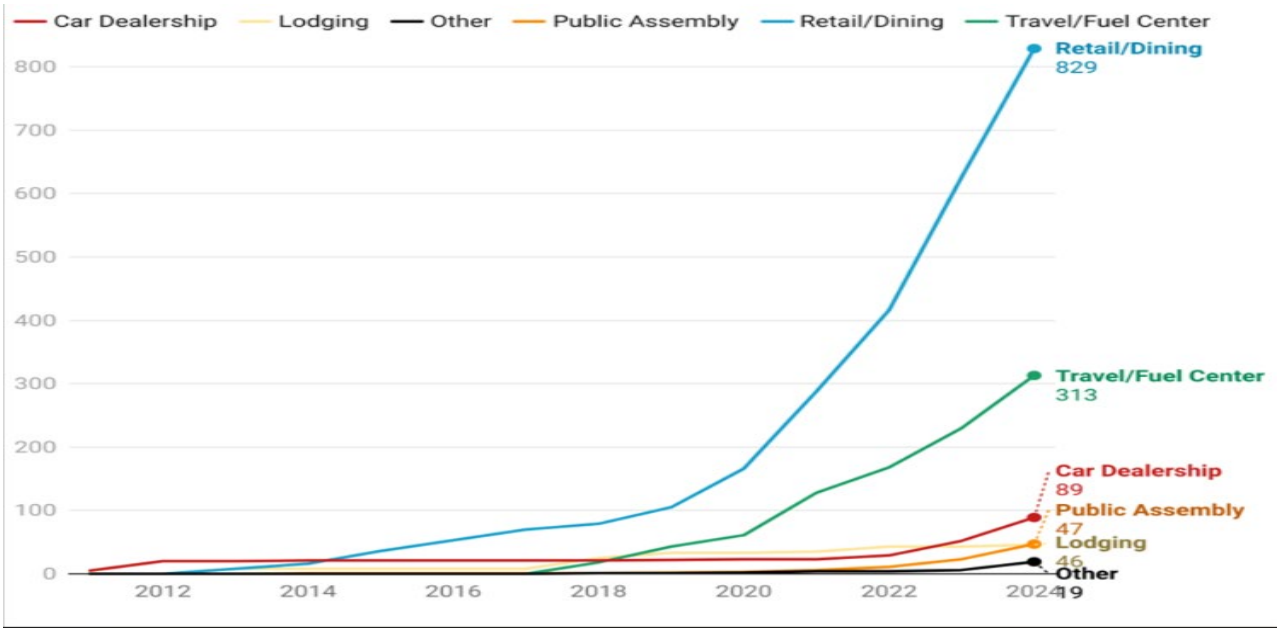


Figure 9: Site locations of direct current (DC) fast-charging ports across North Carolina, 2011–2024

Cost Architecture and Sensitivity

Across industry sources (DOE EVSE Cost Report, EPRI, Qmerit), installation costs for non-residential charging stations cluster around four primary components [2, 29, 37]:

Cost Driver	Total Share	Notes
Labor	55–60 %	Electrical and trenching dominate
Materials	30–35 %	Chargers, conduit, pedestals
Permits + Tax	≈10 %	Local fees, inspection, and zoning
Trenching Adjustment	+ up to 25 %	When conduit > 50 ft from transformer

Labor accounts for the majority of total project expense, especially for sites requiring extensive conduit runs or service upgrades. A simplified relationship can be expressed as

$$C_{install} = C_{base} \times (1 + f_{trench})$$

where $f_{trench} = 0.25$ for long-distance trenching. This formulation highlights why site layout and electrical proximity often outweigh hardware cost when comparing alternatives [2, 37].

The supporting dataset indicates that civil work typically ranges from $36 000 – $104 000, electrical work from $356 000 – $952 000, and mechanical work from $16 000 – $1.67 million, depending on scope [37]. These figures encompass not only charger hardware but also trenching, conduit, foundations, bollards, signage, and accessibility modifications. Because of the expense scales with distance and site preparation, early coordination with utilities is critical [11]. A service-capacity study can reveal whether a transformer upgrade or secondary panel is required—both high-impact cost items. Failing to confirm these parameters early can inflate budgets by tens of thousands of dollars.

Standardizing designs and contractor networks helps reduce soft costs tied to permitting and mobilization [11]. Re-using a tested site template—such as a four-port layout with pre-

approved conduit paths—can cut total installation time by 20 % or more [11]. The same logic underpins the NEVI program's push for uniform, 150-kW-rated sites with consistent accessibility and payment standards [11].

Capital Requirements by Charger Type

At the site level, the financial footprint of EV infrastructure varies sharply by charger class. Level 2 chargers (208–240 V) used in retail or workplace settings cost roughly $14 000 – $60 000 per site, averaging about $36 000 for a four-port installation. DC fast chargers, which operate at 480 V and deliver 50 – 350 kW, typically range from $100 000 – $170 000, with complex sites reaching $500 000 – $1 million when heavy electrical work or transformer relocation is required [2, 22].
A useful comparative metric is *cost per port*:

Cost per Port = C_{site} / N_{ports}

Small convenience-store pilots installing four Level 2 ports thus average about $9 000 per port, while highway DCFC sites with two 150-kW chargers can exceed $250 000 per port. This stark difference illustrates why corridor projects depend on federal subsidies and volume efficiency—adding more chargers per site can materially reduce cost per connection through shared permitting, trenching, and grid interconnection [22, 11].

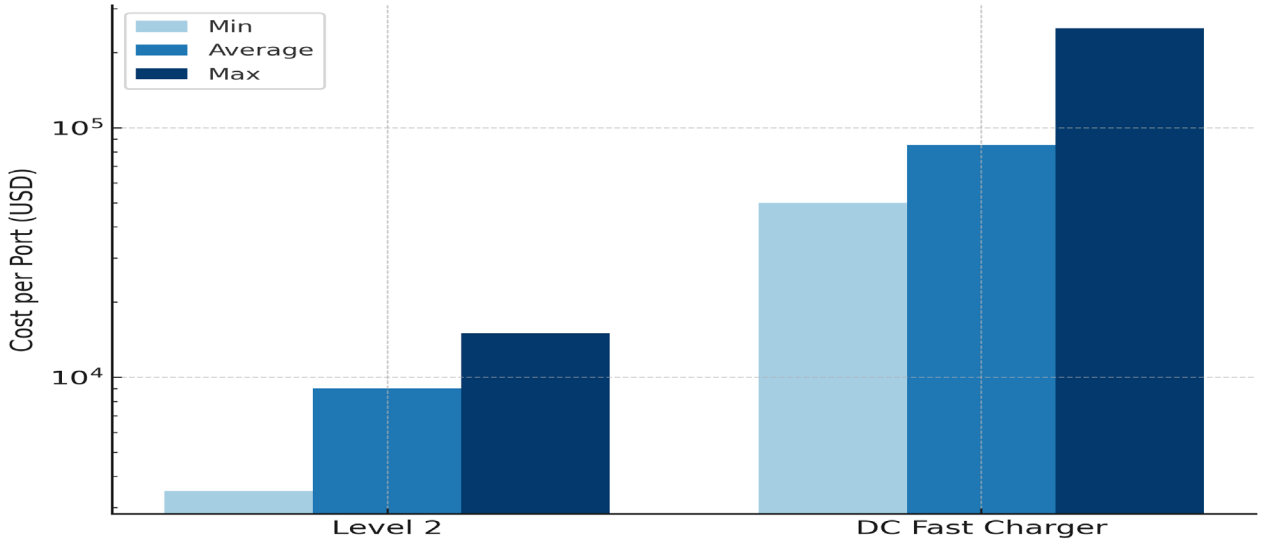


Figure 10: Comparison of Level 2 and DC Fast Charging cost per port

Implementation and Funding Considerations

Effective deployment requires both technical readiness and financial leverage. A typical implementation roadmap begins with pilot installations of small Level 2 packs (three to five sites, $20 000 – $60 000 each) to gauge utilization and user behavior. In parallel, developers engage utilities for service studies at 10 or more candidate DCFC sites [11]. Scaling then proceeds over the next 12–24 months, combining NEVI cost-share grants (covering up to 80 % of eligible expenses) with utility rebates and potential tax credits [11, 17, 35]. Mature networks integrate battery storage and demand-management systems to smooth peak-load costs—reducing exposure to demand-charge penalties that can erode profitability [16].

Beyond direct charger revenue, co-location with retail amenities offers an important secondary benefit: customers dwell for 20–45 minutes, creating cross-sales that help offset slower payback on charging hardware [34, 21]. For this reason, convenience-store forecourts and

shopping-center parking lots are prime near-term expansion zones, while dedicated highway travel centers remain capital-intensive anchor projects best suited to public-private partnerships [7, 11].

In sum, North Carolina's EV-charging build-out rests on two intertwined foundations: technical feasibility and capital discipline. The hardware itself represents only a fraction of total project cost; installation logistics, labor, and electrical proximity drive the rest [2, 37]. Managing these variables—through early utility coordination, standardized designs, and strategic use of federal incentives—determines whether infrastructure expansion remains affordable and scalable [11, 22]. The resulting network will ultimately shape not only how fast the state electrifies but also how efficiently its charging corridors can operate within the broader financial framework of the transition to electric mobility.

Financial Analysis

The financial feasibility of North Carolina's fast-charging network can be expressed not only through revenue or utilization metrics but also through a structured optimization framework that formalizes how limited public and private funds are deployed [9, 11, 26]. For Fiscal Year 2026, the North Carolina Department of Transportation (NCDOT) anticipates $25.08 million in federal NEVI funding, matched by roughly $6.27 million in non-federal capital [28]. The goal is straightforward: to determine the maximum number of NEVI-compliant fast-charging sites that can be built given these fiscal constraints and the program's 80/20 cost-share rule [11, 35].

Name	Value	Unit
NEVI_Public_Budget_FY26	$ 25,084,592	$
Total_Site_Cost_DCFC_4port	$ 400,000	$/site
Federal_Share	0.8	
Private_Match_Pool	$ 6,271,148	$
EVs_per_Site	120	EVs/site
Ports_per_Site	4	ports/site

Figure 11: Fiscal Parameters for NEVI-Funded DC Fast-Charging Site Optimization (FY2026)

At its core, this is a linear integer optimization problem defined as follows [9]:

Maximize $Z = x$

Subject to:

$c^{pub} \times x \leq B^{pub}$

$c^{priv} \times x \leq B^{priv}$

$x \geq 0$ and integer

Where:

x = number of DC fast-charging (DCFC) sites to be funded

C^{site} = total installed cost per site (≈ $400,000)

α = federal cost-share rate (0.80)

B^{pub} = available federal (public) budget (≈ $25,084,592)

B^{priv} = available private-match pool (≈ $6,271,148)

$c^{pub} = \alpha \times C^{site} = \alpha \times C^{site}$ = $320,000 (federal cost per site)

$c^{priv} = (1-\alpha) \times C^{site} = (1 - \alpha) \times C^{site}$ = $80,000 (private cost per site)

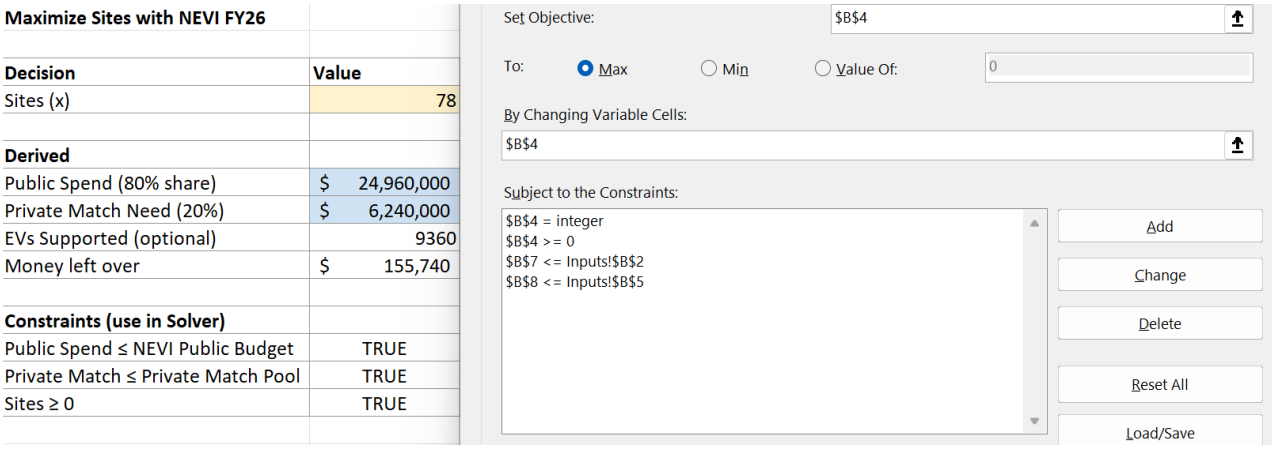
The model's optimal integer solution is given by:

$x^* = \min(\text{floor}(B^{pub} / c^{pub}), \text{floor}(B^{priv} / c^{priv}))$

Substituting the known parameters yields:

$x^* = \min(\text{floor}(25{,}084{,}592 / 320{,}000), \text{floor}(6{,}271{,}148 / 80{,}000))$

$x^* = 78$



Maximize Sites with NEVI FY26

Decision	Value
Sites (x)	78
Derived	
Public Spend (80% share)	$ 24,960,000
Private Match Need (20%)	$ 6,240,000
EVs Supported (optional)	9360
Money left over	$ 155,740
Constraints (use in Solver)	
Public Spend ≤ NEVI Public Budget	TRUE
Private Match ≤ Private Match Pool	TRUE
Sites ≥ 0	TRUE

Figure 12: Solver Optimization Model for NEVI-Funded DC Fast-Charging Sites (FY2026)

Thus, North Carolina can finance 78 standard four-port DCFC sites during FY-2026, fully utilizing both public and private budgets with a small surplus of about $155,000 combined. Each site supports an estimated 120 EVs, yielding potential coverage for approximately 9,360 vehicles statewide [26, 28].

The optimization highlights the leverage power of the NEVI formula: every private dollar mobilizes four additional public dollars, producing a 4:1 investment multiplier [11, 35]. More importantly, it formalizes fiscal boundaries—defining precisely where rising costs or limited private participation begin to cap achievable coverage [11, 28].

Sensitivity tests illustrate this dynamic vividly. If average site costs climb to $450,000, the feasible count drops to about 70 sites; if efficiencies reduce costs to $350,000, the count rises to roughly 90 [2, 22]. Likewise, if private partners can contribute only $4 million, the binding constraint shifts to the private side, limiting deployment to 50 sites even though federal funds remain available [11]. In this way, the model serves as both a planning instrument and an early-warning system for fiscal imbalance [9, 26].

Beyond the arithmetic, the optimization encapsulates the strategic logic of NEVI's funding design. Federal dollars absorb upfront capital risk, ensuring corridor coverage even in low-traffic regions, while private co-investment ensures local ownership and operational accountability [11, 19, 35]. Together, they form a financially balanced mechanism that advances statewide electrification without overextending public budgets [28].

From a policy standpoint, this quantitative model complements the earlier break-even analysis: with NEVI's 80 % subsidy, stations become viable around seven sessions per day. As utilization rises alongside BEV adoption, these sites will naturally progress toward self-sustaining economics—precisely the equilibrium point the model anticipates [12, 16, 20].

Ultimately, fiscal optimization is not an abstract exercise; it is the blueprint for scaling equitable, cost-effective charging coverage. By FY 2026, North Carolina's disciplined use of NEVI funds can translate into 78 corridor-ready sites—312 DC fast ports—positioning the state for both financial efficiency and long-term sustainability in its EV infrastructure rollout [11, 26, 28, 35].

Conclusion

North Carolina's EV transition is well underway, and the story that emerges from this analysis is one of strong momentum paired with strategic growing pains. The state's BEV fleet has expanded at an extraordinary pace, but public-charging development hasn't kept up—creating

pockets of strain in urban hubs, long gaps along corridors, and accessibility challenges in rural and disadvantaged regions [22]. Working through the descriptive, diagnostic, predictive, and prescriptive phases shows how all the pieces fit together: demand is climbing, infrastructure is uneven [24], financial viability hinges on utilization and subsidies [11, 20], and smart siting decisions determine whether investments deliver real value.

A few takeaways rise to the top. First, the BEV-to-charger imbalance is the anchor problem behind most infrastructure bottlenecks [22], and the state will need to add thousands of ports by 2028 just to hit national benchmarks [24]. Second, public charging will not succeed on hardware alone—labor, trenching distance, electrical proximity, and tariff design are just as influential as the charger rating itself [2, 16]. Third, NEVI's 80/20 funding model is doing exactly what it was designed to do: accelerating build-out while keeping economics feasible until the EV fleet grows large enough to support stations without subsidies. Finally, the prescriptive modeling makes one thing clear: carefully targeted deployment produces dramatically better coverage and utilization outcomes than a uniform, one-size-fits-all approach [9].

This work opens the door to several promising extensions. One natural next step is to expand the simulation component—especially through Monte Carlo analysis—to quantify cost and utilization uncertainty at the site level. The Cary, NC storage–fast-charging study provides a strong starting point, but similar simulations could be run statewide, across multiple corridor profiles, or with different tariff and incentive structures. Likewise, the optimization framework introduced in the finance section could be broadened into a full location-allocation model that balances cost, practical infrastructure access for all regions, grid constraints, and expected demand under various adoption scenarios [9, 24]. Together, these methods would give planners a more resilient set of strategies for navigating cost volatility, adoption uncertainty, and shifting policy environments.

In short, North Carolina has the momentum, the funding, and the analytical tools to build a charging network that matches its rapidly growing EV market [26, 24]. The challenge—and the opportunity—lies in continuing to refine the models that guide these investments, making sure every dollar works harder, every corridor stays connected, and every community has a place in the transition. This paper lays the foundation; the next phase of research will help ensure the system we build is not only functional today, but adaptable for the decade ahead.

References

1. Abbott, D. (2014). *Applied predictive analytics: Principles and techniques for the professional data analyst* [e-book]. Wiley. https://www.vitalsource.com/products/applied-predictive-analytics-dean-abbott-v9781118727690
2. Advanced Energy, & City of Raleigh. (2022). *Raleigh's EV-Ready Playbook*. Advanced Energy. https://www.advancedenergy.org/wp-content/uploads/imported-files/City-of-Raleigh-Playbook-Final-Version.pdf
3. Alternative Fuels Data Center. (2015). *Costs associated with non-residential electric vehicle supply equipment: Factors to consider in the implementation of electric vehicle charging stations*. U.S. Department of Energy. https://afdc.energy.gov/files/u/publication/evse_cost_report_2015.pdf
4. Alternative Fuels Data Center. (n.d.-a). *Charging electric vehicles at home*. U.S. Department of Energy. https://afdc.energy.gov/fuels/electricity-charging-home
5. Alternative Fuels Data Center. (n.d.-b). *North Carolina alternative fuels and charging station information*. U.S. Department of Energy. https://afdc.energy.gov/states/nc
6. Broad Reach Construction. (n.d.). *EV charging infrastructure services*. https://broadreachconstruction.com/services/ev-charging
7. ChargePoint. (n.d.). *EV charging solutions for businesses, fleets, and drivers*. https://www.chargepoint.com
8. CSNews. (2024, September 12). *Love's, Pilot receive contracts for EV charging stations in North Carolina*. *Convenience Store News*. https://csnews.com/loves-pilot-receive-contracts-ev-charging-stations-north-carolina
9. Electrify America. (n.d.). *Public DC fast-charging network*. https://www.electrifyamerica.com
10. Evans, J. R. (2021). Business analytics: Methods, models, and decisions (3rd ed.). Pearson Education, Inc.
11. EVgo. (n.d.). EVgo fast-charging network. https://www.evgo.com
12. Federal Highway Administration. (2024, June 11). National Electric Vehicle Infrastructure (NEVI) Formula Program guidance. U.S. Department of Transportation. https://www.fhwa.dot.gov/environment/alternative_fuel_corridors/nominations/90d_nevi_formula_program
13. Fröde, P., Lee, M., & Sahdev, S. (2023, October). Can public EV fast-charging stations be profitable in the United States? McKinsey & Company. https://www.mckinsey.com/features/mckinsey-center-for-future-mobility/our-insights/can-public-ev-fast-charging-stations-be-profitable-in-the-united-states
14. Galit Shmueli, Bruce, P. C., Stephens, M. L., Muralidhara, A., & Patel, N. R. (2023). *Machine learning for business analytics*. Wiley.
15. Generation180. (2022). *North Carolina drives electric 2022: Strong consumer demand highlights electric vehicles as an economic driver*. Generation180. https://generation180.org/wp-content/uploads/North-Carolina-Drives-Electric.pdf
16. Governor's Office of North Carolina. (2024, September 11). *Governor Cooper announces first awards for electric vehicle charging station program*. https://governor.nc.gov/news/press-releases/2024/09/11/governor-cooper-announces-first-awards-electric-vehicle-charging-station-program
17. Haris, B. (2024). *Rate design for EV fast charging stations: A case study from Duke Energy Progress* [White paper]. UNC Kenan-Flagler Business School. https://www.kenan-flagler.unc.edu/wp-content/uploads/2024/05/White-Paper-HP-and-EV-Charging.pdf

18. Internal Revenue Service. (2025). *Credit for alternative fuel vehicle refueling property (Form 8911 instructions)*. U.S. Department of the Treasury. https://www.irs.gov/forms-pubs/about-form-8911
19. International Council on Clean Transportation. (n.d.). *EV charging infrastructure research and policy*. https://theicct.org
20. Joint Office of Energy & Transportation. (2024). *Community charging: Emerging multifamily, curbside, and community hub strategies*. https://driveelectric.gov/files/community-emobility-charging.pdf
21. McKinsey & Company. (2022, April 18). *Building the electric-vehicle charging infrastructure America needs*. https://www.mckinsey.com/industries/public-sector/our-insights/building-the-electric-vehicle-charging-infrastructure-america-needs
22. National Association of Convenience Stores. (n.d.). *U.S. convenience store industry count and statistics*. https://www.convenience.org
23. National Renewable Energy Laboratory. (2023). *The 2030 national charging network: Estimating U.S. light-duty demand for electric vehicle charging infrastructure*. U.S. Department of Energy. https://www.nrel.gov/docs/fy23osti/85332.pdf
24. NC OneMap. (2025). *Gas stations in North Carolina* [Geospatial dataset]. North Carolina Department of Information Technology. https://www.nconemap.gov
25. North Carolina Department of Transportation. (n.d.-a). *Electric vehicle deployment plan*. https://www.ncdot.gov/initiatives-policies/environmental/climate-change/NEVI/Documents/ncdot-electric-vehicle-deployment-plan.pdf
26. North Carolina Department of Transportation. (n.d.-b). *NEVI Phase 1 basics (fact sheet)*. https://www.ncdot.gov/initiatives-policies/environmental/climate-change/NEVI/Documents/nevi-basics-fact-sheet-phase-1-v2-final.pdf
27. North Carolina Department of Transportation. (2023). *Growing North Carolina's zero-emission vehicle market: Infrastructure needs assessment*. NCDOT. https://www.ncdot.gov/initiatives-policies/environmental/climate-change/Documents/ncdot-zev-infrastructure-needs-assessment.pdf
28. North Carolina Department of Transportation. (2024). *2024 FHWA plan update for electric vehicle deployment*.
29. North Carolina Department of Transportation. (2025). *North Carolina electric vehicle infrastructure deployment plan (2025 update)*. https://www.ncdot.gov/initiatives-policies/environmental/climate-change/NEVI/Documents/nevi-state-plan-update-2025.pdf
30. Qmerit. (2024, February). *Installing EV charging in multifamily properties*. Qmerit. https://qmerit.com/wp-content/uploads/2024/02/Installing-EV-Charging-Multifamily-0224.pdf
31. Qmerit. (n.d.). *EV charging installation and electrification solutions*. https://qmerit.com
32. Rocky Mountain Institute. (n.d.). *EV charging and grid integration research*. https://rmi.org
33. SAS Institute Inc. (2025, March). *Statistics you need to know for machine learning* [Online course]. SAS Learn. https://learn.sas.com/
34. U.S. Department of Energy. (2024, November 8). *National EV Charging Network*. https://www.energy.gov/topics/national-ev-charging-network
35. U.S. Department of Energy, Vehicle Technologies Office. (2023, December 4). *FOTW #1319: EV charging at paid DC fast charging stations average 42 minutes per session*. https://www.energy.gov/eere/vehicles/articles/fotw-1319-december-4-2023-ev-charging-paid-dc-fast-charging-stations-average

36. U.S. Department of Transportation. (2025, January 31). *Federal funding programs – NEVI (Rural EV toolkit)*. https://www.transportation.gov/rural/ev/toolkit/ev-infrastructure-funding-and-financing/federal-funding-programs
37. U.S. Department of Transportation, Federal Highway Administration. (2025, March 13). *Highway Statistics 2023: Table VM-1—Annual vehicle distance traveled in miles and related data, by highway category and vehicle type*. https://www.fhwa.dot.gov/policyinformation/statistics/2023/vm1.cfm
38. WattLogic. (n.d.). *Commercial EV charging cost guide*. https://wattlogic.com
39. Yes Solar Solutions. (n.d.). *EV charging installations and solar integration*. https://yessolarsolutions.com